

Role of Height and Arm Span in Morphological Optimization of Male Archery Athletes

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ABSTRACT

The study was envisaged with the view to find out the effect of height and arm span on archery performance scores. In the present cross-sectional study, the data of 215 male junior and cadet archers (146 recurve and 69 compound archers), aged below 20 years was collected by purposive sampling. The data was collected during a trail event at Sports Authority of India, North Regional Centre, Sonapat. For recording the archery performance scores, archers participated in two scorings rounds. Data for anthropometric parameters, height and arm span were measured before scoring rounds. In recurve archers linear model ($\beta=0.717$, $p=0.010$) of height predict the scores significantly and the variation in height explains 4.6% of the variation in the archery performance scores. However, in compound archers height predict the scores non-significantly. Among both compound ($\beta=-0.049$, $p=0.007$) and recurve archers ($\beta=-0.049$, $p=0.004$) quadratic model of arm span predict the scores significantly. The variation in arm span explains 13.8% and 7.4% of the variation in the archery performance scores among compound and recurve archers respectively. The findings of the present study can be used to define the role of height and arm span along with other parameters in designing criteria for fresh talent induction in archery.

KEY WORDS: Talent identification, Morphological optimization, Archery, Height, Arm Span

INTRODUCTION

The study of body structure and composition with reference to sports performance always remains the area of interest. The individual variations in the body shape and size contributes to the variations in the performing skills (Peters, 1986; Okely, 2004). The

established of particular successful body shape and composition as a result of selective pressure of success over the consecutive generations and as adaption to contemporary training demands can be termed as morphological optimization (Norton, Olds, Olive & Craig, 1996;

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Rakovac, Smoljanović, Bojanić, Hannafin, Hren & Thomas, 2011; Alacid, Marfell-Jones, Muyor, López-Miñarro & Martínez, 2015). The role of certain anthropometric and body composition parameters towards the sports performance is well established (Norton & Olds, 1996; Ackland, Bloomfield & Elliot, 2009). In 2008, Lozovina and Lozovina suggested to make the primary selection of athletes at early ages on the basis of their morphological features, followed by secondary round of selection at later ages on the basis of psychomotor abilities.

Archery is a sport which involves shooting multiple arrows repeatedly on the same target. To achieve high scores, archers have to follow the correct shooting technique, consistently and very precisely. During competition an archer has to shoot 72 arrows on the target placed at the distance of 70m (in recurve events) and 50m (in compound events). Owing to the high level of precision required for archery, it become imperative to minimizing postural sway and follow proper technique during the aiming phase that allows for higher repeatability, consistency and reduce fatigue. Physical structure of the athlete greatly affects the postural sway and ability of the archer to apply the proper technique during shooting (Musa, Taha, Abdul Majeed, & Abdullah, 2019; Taha et al., 2017; Kim, Kwon, Chung, Kim, Eom, Jun & Park BK, 2012).

Height and arm span are two most considered anthropometric measurements for studies assessing effect of body structure on archery performance (Shams, Matinhomate & Perri, 2015; Pelana and Winata, 2018). Height become important as it is one of the factors in deciding the trajectory of arrow during the flight. Trajectory of arrow affects the duration for which arrow remains in air and face the effect of environmental factors. It has been found from number of studies that taller height affects the archery performance positively (Lau, Ghafar, Hashim, & Zulkifli, 2020; Musa, Taha, Abdul Majeed, & Abdullah, 2019; Humaid, 2014). However, in some studies it has been observed that the individuals with taller height produce more sway in their static posture (Alonso AC et al, 2015; Tabrizi, Abbasi & Sarvestani, 2013), that may impact the archery performance negatively (Taha et al., 2017).

Another key anthropometric parameter that significantly impact archery performance is the arm span. It has biomechanical importance in archery technique because the draw length of the archer completely depends upon the arm span. The studies conducted to look into the role of archery arm span in archery performance presented inconsistent results (Spratford and Campbell, 2017; Taha et al., 2017). So, it becomes very important to investigate the direct relation between the height and arm span,

and archery performance scores.

There is complete gap in information providing direct evidence showing the type of relationship between the anthropometric parameters with archery performance scores among Indian archers. Understanding such relations may further help in identifying role of anthropometric parameters in fresh talent detection for archery. The current study was envisaged with the view to find the effect of height and arm span on archery performance scores. This will put forward the type and extent of relationship that exist between the anthropometric measurements, height and arm span, and archery performance score. None of the earlier studies on the present topic with reference to Indian archers involve such a large participation. All the archers that participated in the study, shoot at the same time during the two scoring rounds. This reduced the variations in the scores that could have been caused due to environmental and other confounding factors. Thus, the study will provide comprehensive information about the effect of height and arm span on archery performance scores.

METHODOLOGY

Sampling and data collection

In the present cross-sectional study, the data of 215 male junior and cadet archers (146 recurve and 69 compound archers), aged below 20 years was collected. The sample was collected by purposive

sampling technique during the trail event at Sports Authority of India, North Regional Center, Sonapat. Data was collected for general information, archery performance scores and anthropometric parameters. Each participant was informed about the methodology and purpose of the study and a written consent was obtained from each subject during the collection of data.

Archery performance Scoring

For recording the archery performance scores, all the archers participated in two scorings, each scoring has 6 rounds and in each round 6 arrows were shoot. For each arrow scores can vary between 0 (when arrow do not hit target at all) and 10 (when arrow hit at the centre of target). The total maximum score for both rounds was 720. All the archers shoot at same time during scoring to rule out the role of the any confounding environmental factors in variation of the archery performance score. The archers that had scored distinctly low were excluded from the study.

Anthropometric parameters

Data for anthropometric parameters, height and arm span were measured for each subject before scoring. Anthropometric measurements were conducted using the ISAK protocol (Arthur, Michael, Timothy & Hans, 2011). Height was measured with the help of a seca stadiometer (seca 284) and the

measurements were recorded to the nearest 1 millimetre. Arm span was measured using Harpenden anthropometer and the measurement was recorded to the nearest 1 millimetre.

Statistics

Kolmogorov-Smirnov test was performed to evaluate the data distribution normality. Descriptive statistics including the mean, median, standard deviation (SD), minimum (Min.), maximum (Max.) and percentiles were calculated for the arm span and height of the compound and recurve archers. For the purpose of the analysis, the archers were divided into four groups on the basis of percentile of height and arm span (upto 25th percentile, 26th to 50th percentile, 51st to 75th percentile and 76th to 100th percentile). The type and significance of relation between the archery performance score, and height and arm span were analyzed by curve linear regression analysis. In curve linear regression analysis score is considered as

dependent variable whereas height and arm span are independent variable. All the statistical analysis was performed using SPSS version 21 for windows (IBM Corporation, Armonk, NY, USA).

RESULTS

In the present study mean age of compound and recurve archers was 17.83 years and 17.43 years, respectively. Descriptive statistics for height and arm span for compound and recurve archers are provided in table 1. Standard deviation (SD), minimum (Min.), maximum (Max.) and values at 25th, 50th and 75th percentiles explain the variation and distribution of height and arm span in compound and recurve archers.

The mean scores of the compound and recurve archers according to percentile height distributions are provided in table 2 and table 3 respectively and the graphical illustration for the same is displayed in figure 1a and 1b respectively.

Table 1: Descriptive statistics of height and arm span among compound and recurve archers.

Variables	Event	N	Mean	Median	SD	Min.	Max.	Percentile		
								25 th	50 th	75 th
Height	Compound	69	172.05	172.60	6.37	158.19	184.60	168.1	172.60	176.60
	Recurve	146	171.93	172.10	6.88	151.50	189.40	167.67	172.10	177.00
	Total	215	171.97	172.40	6.71	151.50	189.40	167.9	172.40	176.70
Arm Span	Compound	69	177.43	176.90	7.20	155.10	195.20	172.60	176.90	183.20
	Recurve	146	177.72	177.35	8.66	157.00	201.00	172.00	177.35	183.60
	Total	215	177.63	177.10	8.20	155.10	201.00	172.40	177.10	183.50

It can be observed in compound archers that mean scores increase continuously with increase in height from 326.13 scores for 1st - 25th percentile group to 335.00 scores in 76th - 100th percentile group. On the other hand, in recurve

archers mean scores drop a little from 299.84 for bottom 1st - 25th percentile group to 298.92 for 26th - 50th percentile group and, then increase continuously to 309.27 till 76th - 100th height percentile group.

Table 2: Scores according to percentile height distribution in compound archers

Height Percentiles	Compound			
	N	Height		Scores
		Mean	Range	Mean $\pm$ SD
1 st - 25 th	19	164.18	158.19-168.10	326.13 $\pm$ 10.98
26 th - 50 th	17	170.46	168.20-172.60	327.82 $\pm$ 16.08
51 st - 75 th	16	174.57	172.70-176.50	331.40 $\pm$ 17.32
76 th -100 th	17	180.08	176.70-184.60	335.00 $\pm$ 12.71

Table 3: Scores according to percentile height distribution in recurve archers

Height Percentiles	Compound			
	N	Height		Scores
		Mean	Range	Mean $\pm$ SD
1 st - 25 th	36	163.12	151.50-167.60	299.84 $\pm$ 25.85
26 th - 50 th	38	169.77	167.70-172.10	298.92 $\pm$ 22.99
51 st - 75 th	37	174.60	172.40-177.00	307.56 $\pm$ 19.71
76 th -100 th	35	180.53	177.20-189.40	309.27 $\pm$ 22.62

The mean scores of the compound and recurve archers according to percentile arm span distributions are provided in table 4 and table 5 respectively and the graphical illustration for the same is displayed in figure 1c and 1d respectively. It can be observed in compound archers that the mean scores

increase continuously with increase in arm span from 323.38 scores for 1st - 25th percentile group to 334.38 scores in 51st - 75th percentile group and thereafter scores decrease to 331.34 scores for 76th-100th percentile group. Similarly, in recurve archers the mean scores increase continuously with increase in arm span

from 297.67 scores for 1st - 25th percentile group to 311.05 scores in 51st - 75th percentile group and thereafter scores

decrease to 303.67 scores for 76th - 100th percentile group.

Table 4: Scores according to percentile arm span distribution in compound archers

Arm span Percentiles	Compound			
	N	Arm Span		Scores
		Mean	Range	Mean $\pm$ SD
1 st - 25 th	17	168.54	155.10-172.40	323.38 $\pm$ 15.09
26 th - 50 th	18	174.88	172.80-176.90	330.50 $\pm$ 10.28
51 st - 75 th	18	180.20	177.30-183.20	334.38 $\pm$ 15.77
76 th - 100 th	16	186.65	183.50-195.20	331.34 $\pm$ 15.10

Table 5: Scores according to percentile arm span distribution in recurve archers

Arm span Percentiles	Compound			
	N	Arm Span		Scores
		Mean	Range	Mean $\pm$ SD
1 st - 25 th	36	166.82	157.00-172.00	297.67 $\pm$ 26.94
26 th - 50 th	38	174.69	172.20-177.10	302.33 $\pm$ 23.46
51 st - 75 th	37	181.02	177.60-183.60	311.05 $\pm$ 18.36
76 th - 100 th	35	188.80	184.30-201.00	303.67 $\pm$ 21.80

Figure 1: Graphs depicting mean scores and standard deviations of compound and recurve archers according to increasing height and arm span percentiles.

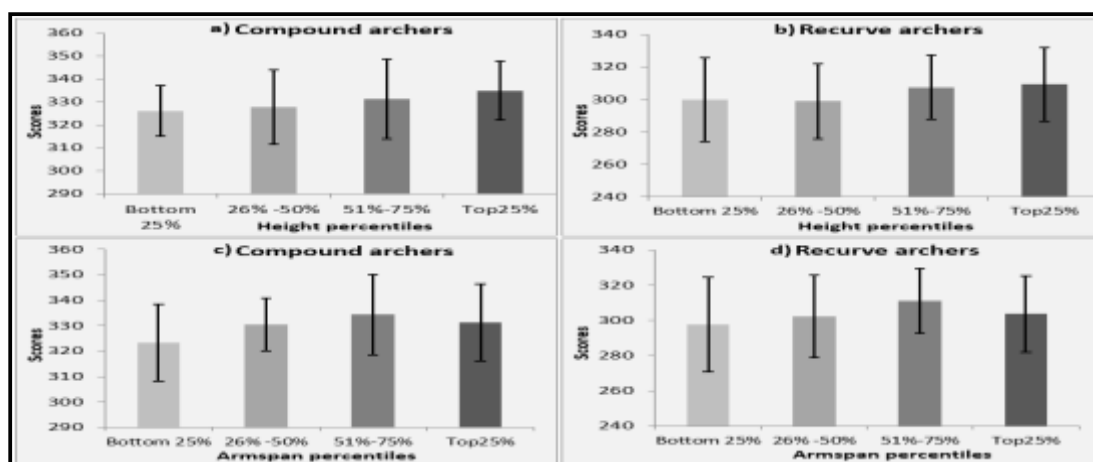


Table 6: Curve linear regression analysis comparing models for relationship between score and height among compound and recurve archers

Event	Variable	Model	Beta (β)	R	R Square	<i>F</i>	<i>p</i>
Compound	Height	Linear	0.540	0.238	0.057	4.019	0.049
		Quadratic	-0.035	0.265	0.070	2.497	0.090
		Change	-	-	0.014	0.977	0.327
Recurve		Linear	0.717	0.213	0.046	6.870	0.010*
		Quadratic	-0.051	0.262	0.069	5.268	0.006*
		Change	-	-	0.023	3.544	0.062

*Statistically significant

Curve linear regression analysis, comparing the linear and quadratic models of height for their ability to predict the variation in archery performance scores of compound and recurve archers are presented in table 6 and graphical illustration of the same is provided in figure 2a and 2b. In compound archers, both linear ($\beta = 0.540$, $p = 0.049$) and quadratic models ($\beta = -0.035$, $p = 0.090$) of height predict the scores non-significantly. It indicated that both linear and quadratic models of height play non-significant role in variation of the archery performance

scores of compound archers. In recurve archers, both linear model ($\beta = 0.717$, $p = 0.010$) as well as quadratic model ($\beta = -0.051$, $p = 0.006$) of height predict the scores significantly. However, R^2 change from linear to quadratic model is non-significant ($F = 3.544$, $p = 0.062$), indicating that among recurve archers linear model for height play dominant role in variation of archery performance scores. Moreover, the R^2 value of 0.046 depicts that the linear model for height explain 4.6 % of variation in archery performance scores.

Table 7: Curve linear regression analysis comparing models for relationship between score and arm span among compound and recurve archers

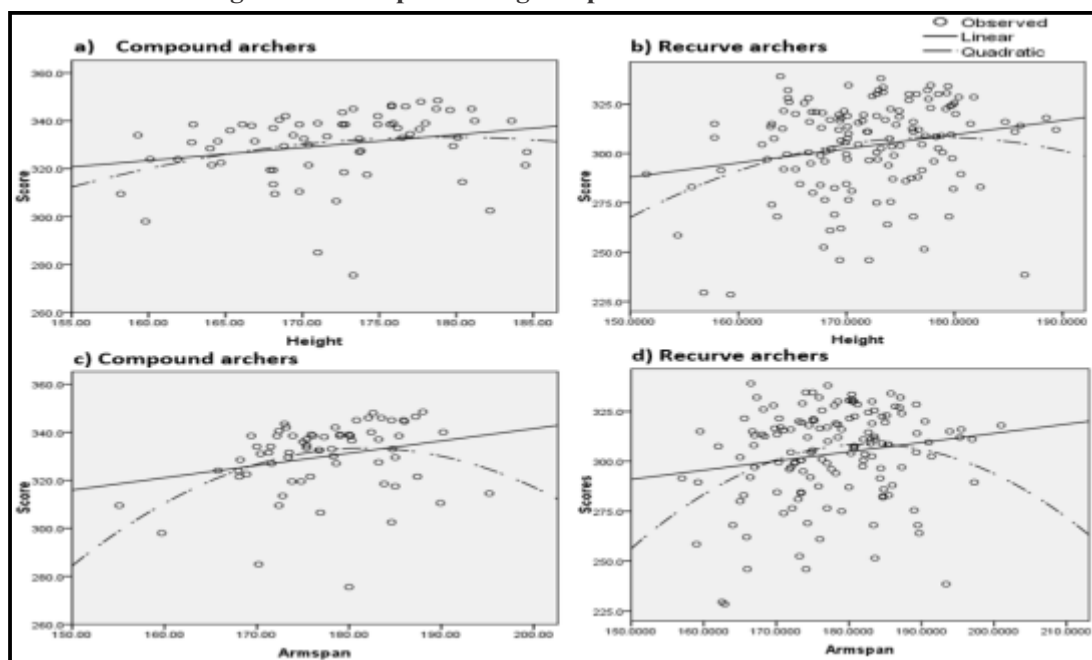
Event	Variable	Model	Beta (β)	R	R Square	<i>F</i>	<i>p</i>
Compound	Arm span	Linear	0.512	0.255	0.065	4.658	0.034
		Quadratic	-0.049	0.371	0.138	5.282	0.007*
		Change	-	-	0.073	5.588	0.021*
Recurve		Linear	0.461	0.173	0.030	4.428	0.037
		Quadratic	-0.049	0.272	0.074	5.702	0.004*
		Change	-	-	0.044	6.798	0.010*

*Statistically significant

Curve linear regression analysis, comparing the linear and quadratic models of arm span for their ability to predict the variation in archery performance scores of compound and recurve archers are presented in table 7 and graphical illustration of the same is provided in figure 2c and 2d. Among compound archers, both linear model ($\beta=0.512$, $p=0.034$) as well as quadratic models ($\beta=-0.049$, $p=0.007$) of arm span predict the scores significantly. Also, R^2 change from linear to quadratic model of arm span is significant ($F=5.588$, $p=0.021$), indicating that among compound archers quadratic model for arm span play significant role in variation of the archery performance scores ($\beta=-0.049$, $p=0.007$). The R^2 value of 0.138

depicts that the quadratic model for arm span explain 13.8 % of variation in archery performance scores of compound archers. In recurve archers, both linear model ($\beta=0.461$, $p=0.037$) as well as quadratic model ($\beta=-0.049$, $p=0.004$) of arm span predict the archery performance scores significantly. Also, R^2 change from linear to quadratic model of arm span is significant ($F=6.798$, $p=0.010$), indicating that among compound archers quadratic model for arm span play significant role in variation of archery performance scores. Moreover, the R^2 value of 0.074 depicts that the quadratic model for arm span explain 7.4% of variation in archery performance scores of recurve archers.

Figure 2: Curve linear regression analysis graphs for linear and quadratic models of height and arm span among compound and recurve archers.



DISCUSSION

The quest for the medals at the international platforms like Olympics has increased over time. This has provided impetus to select the talent, that could increase the likelihood of success at senior levels. In 2013, Pankhurst and Collins suggested to conduct research for talent identification involving multiple stake holders. However, before identifying the talent, it is important to specify what we are looking for as talent. Anatomical, physiological and psychological features of the human body can impact the sports performance significantly (Mermier, Janot, Parker & Swan, 2000, Bayios, Bergeles, Apostolidis, Noutsos & Koskolou, 2006, Ziv & Lidor, 2009). Also, as the anatomical features cannot be modified to a greater extent after maturation, therefore they may act as the limiting factor in improving the performance after the period of growth and development (Lozovina & Lozovina, 2008). So, it become imperative to look for such relationship in respective sports (Moura, Costa, Oliveira, Júnior, Ritti-Dias & Santos, 2014, Silva, Petroski & Gaya, 2013). Therefore, the present study was conducted to develop a view on effect of the height and arm span on archery performance scores.

The findings of the present study showed that height has a significant impact on archery performance scores in recurve archers. On the other hand,

among compound archers height showed a non-significant effect on archery performance scores. On further analysis of the type of relationship, it was observed that among recurve archers the height has a linear association with the archery performance scores. That is the archery performance scores increases with increase in height. The present study has also found that variation in height explains 4.6% of the variation in the archery performance scores among recurve archers. In 2016, Debnath, Roy, Chatterjee & Dey observed the average height of 170.42 cm among archery male national campers from SAI, Kolkata. In our study, the average height of the male archers was observed to be 171.97 cm. The height in both studies is almost within same range.

In current study, arm span significantly affects the archery performance scores in both compound and recurve archers. On further analyzing the type of relationship, it was observed that among both compound and recurve archers the arm span has a quadratic association with the archery performance scores. This indicates that initially archery performance scores increase with increasing arm span, however with further increase in the arm span the archery performance scores are impacted negatively. Also, the variation in arm span explains 13.8% and 7.4% of the variation in the archery performance scores among compound and recurve archers

respectively. The findings indicated that although a wider arm span is beneficial for archery but exceptionally wider arm span may impact the archery performance negatively.

In 2017, Taha et al. explored the role of various physical characters towards archery performance and observed that a wider arm span and taller height had a negative impact on archery performance. On the other hand, Musa et al., 2019 and Lau et al., 2020 found that the high-performance archers had a wider arm span and taller height as compared to the low performance archers. Our study has found that the archery performance scores increased with increasing with arm span till 50th to 75th arm span percentile group. However, the archery performance scores decreased if arm span increased further. Unlike earlier studies in which comparison was made between high performing and low performing archers, current study has explored a direct relation between the arm span and archery performance scores. Therefore, presented a different view on the type of relationship between the two parameters.

The observations of the present study regarding the relation between the arm span and score can be interpreted with regard to draw length and the torque produced at the distal end of the bow hand in archery. The archers with wider arm span have longer draw length. The wider arm span and hence the longer draw length make it easy for archers to pull the

greater poundage with more convenience. Greater poundage increases draw force, release the arrow at higher velocity, lesser flight time and better scores (Spratford and Campbell, 2017). However, due to the biomechanical limitation very long arms present the strength problems, the torque increases at the distal end of arm (Ackland, Bloomfield, & Elliot, 2009). This increases the sway in the bow hand and impacts the archery scores negatively (Taha et al., 2017).

The physical, physiological and psychological factors are important in fresh talent induction in archery (Kumar, 2021, Kim, Kim, & So, 2015). The findings of the present study can be used to define the role of height and arm span along with other parameters in designing criteria for fresh talent induction in archery. Many earlier studies also suggested including height and arm span measurements as a part of talent induction criteria (Humaid, 2014, Lau, Ghafar, Hashim, & Zulkifli, 2020).

CONCLUSION

The findings of the present study showed that height has a significant linear association with archery performance scores and variation in height explains 4.6% of the variation in the archery performance scores in recurve archers. The arm span significantly quadratic association with archery performance scores in both compound and recurve

archers. And, the variation in arm span explains 13.8% and 7.4% of the variation in the archery performance scores among compound and recurve archers respectively.

The findings of the present study can be used to define the role of height and

arm span along with other parameters in designing criteria for fresh talent induction in archery.

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